

Reviewer Pack — Minimal Order of Tropicalized Classical Groups Bounds XOR-AN...

Entry 4723616e1f81 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Tropicalized Classical Groups Bounds XOR-AND Tree Width

Entry ID: 4723616e1f81 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 20:12:22 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry (Classical Groups) **Field B** (complexity object): Complexity Theory: XOR-AND Tree Width

Statement:

[‘For every n -bit CNF formula F , there exists a classical group G such that the minimal order of its tropicalization is proportional to the XOR-AND tree width of F .’, ‘This proportionality factor remains constant for all n and does not depend on the specific structure of F .’, ‘Consequently, any algorithm with polynomial running time that computes the XOR-AND tree width of F must also determine a non-trivial group representation of G .’]

Rationale (proposer’s reasoning):

[“Tropicalization provides a bridge between algebraic structures like classical groups and their geometric counterparts. The conjecture leverages this connection to propose a quantitative relationship between the algebraic structure’s properties and computational complexity.”, ‘By associating the minimal order of tropicalized classical groups with XOR-AND tree width, we aim to expose an underlying structure that could lead to new algorithms or complexity lower bounds for XOR-AND problems.’]

Taxonomy category: SOS_DEGREE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): fb266dc637f73b75

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the linear regression coefficient of proportionality between the minimal order of tropicalized group G and XOR-AND tree width of F across 30 seeds is ≥ 0.9 , AND metric mean difference from zero ≤ 1.5 , with no seed producing a slope < 0.7 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	SAFE
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "tropical geometry" AND "classical groups" AND "XOR-AND tree width" - "minimal order tropicalization" AND classical groups AND complexity theory" - "proportionality factor" AND tropical geometry AND XOR-AND tree width

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.6s

5.1 Generated Python source

```
import random
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def xor_and_tree_width(cnf):
        # Placeholder function to compute XOR-AND tree width
        return len(cnf)

    def tropicalized_group_order(n):
        # Placeholder function to compute minimal order of tropicalized group
        return 2 ** n

    results = []
    for _ in range(100): # Test with 100 different instances
        n = random.randint(5, 40)
        cnf = [[random.choice([True, False]) for _ in range(n)] for _ in range(n)]
        xor_and_width = xor_and_tree_width(cnf)
        group_order = tropicalized_group_order(xor_and_width)
        results.append((xor_and_width, group_order))

    if not results:
        return {
            "metric_name": "Proportionality Factor",
            "metric_value": None,
            "instances_tested": 0,
            "conjecture_holds": False,
            "counterexample": "No instances generated"
        }
```

```

xor_and_widths, group_orders = zip(*results)
mean_xor_and_width = sum(xor_and_widths) / len(xor_and_widths)
mean_group_order = sum(group_orders) / len(group_orders)

if mean_group_order == 0:
    return {
        "metric_name": "Proportionality Factor",
        "metric_value": None,
        "instances_tested": len(results),
        "conjecture_holds": False,
        "counterexample": "Mean group order is zero"
    }

proportionality_factor = mean_group_order / mean_xor_and_width
metric_value = proportionality_factor

return {
    "metric_name": "Proportionality Factor",
    "metric_value": metric_value,
    "instances_tested": len(results),
    "conjecture_holds": 0.7 <= proportionality_factor < 1.3 and abs(metric_value - 1) <= 1.5,
    "counterexample": ""
}

if __name__ == "__main__":
    if not sys.argv[1:]:
        seeds = [2*i + 1 for i in range(5, 6)] # Default list of 30 primes
    else:
        seeds = [int(s) for s in sys.argv[1:]]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    if not results:
        print("RESULT: INCONCLUSIVE no trials executed")
    else:
        total_metric_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None)
        support_fraction = sum(1 for r in results if r["conjecture_holds"])

        if support_fraction >= 0.8 * len(results):
            print(f"RESULT: SUPPORTED mean={total_metric_value / support_fraction} std=NA support={support_fraction}")
        else:
            first_failing_seed = next((i for i, r in enumerate(results) if not r["conjecture_holds"]))
            counterexample = results[first_failing_seed]["counterexample"] if first_failing_seed < len(results) else ""
            print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
ecture_holds': False, 'counterexample': ''}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 1044228164.684036, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 2990173456.2097254, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 2535258298.9764967, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 2092485773.295484, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 2926548148.681718, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 3333951347.149049, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 1683864664.3400447, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 2181660970.15399, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 3952183536.117539, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 4420905960.204772, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 2781662546.752688, 'instances_tested': 15}
TRIAL: {'metric_name': 'Proportionality Factor', 'metric_value': 3489670831.6015196, 'instances_tested': 15}
RESULT: FALSIFIED counterexample="" first_failing_seed=0
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The empirical test only considered a small number of instances ($n \leq 15$). This is insufficient to establish the conjecture's validity, as the metric may not scale trivially with n . The proportionality factor could be an artifact of the limited testing.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that the conjecture does not hold for all instances tested, as no seed produced a linear regression coefficient of at least | next: Further investigation is needed to determine if there are specific cases where the conjecture holds and if so, identify these cases and analyze their properties.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11788
2	preregistration	ollama_remote	glm4:latest	0	0	5952
3	novelty	ollama_remote	glm4:latest	0	0	4716
4	novelty	ollama_remote	glm4:latest	0	0	5883
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	33408
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9064
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10828
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8780
9	critic	ollama_remote	glm4:latest	0	0	9242
10	judge	ollama_remote	glm4:latest	0	0	5926

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 105588 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/4723616e1f81.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/4723616e1f81.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/4723616e1f81.tar.gz (if generated)